

Ayush Roy

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EDUCATION

State University at New York, Buffalo Aug'2024 – Present
Doctor of Philosophy (Ph.D.), Computer Engineering
Advisor: **Dr. Vishnu Suresh Lokhande**

Jadavpur University, West Bengal Jul'2020 - Aug'2024
Bachelor of Engineering, Electrical Engineering

RESEARCH EXPERIENCE

Lawrence Livermore National Lab, LLNL May'2026 - Aug'2026
[Computational Engineering Research Intern](#)

Indian Institute of Science and Technology, IISc May'2023 - Aug'2023
[Summer Research Intern](#)

SERVICES

TMLR, DMLR, ICLR 2027

PUBLICATIONS

See my [Google Scholar page](#) for all publications, recent updates and citation information. (*: *equal contributions*)

Peer-Reviewed Conference Proceedings

- [C1] A Roy, R Chakraborty, L Varshney, V Lokhande. **Beyond Pooling: Matching for Robust Generalization under Data Heterogeneity.** *AISTATS 2026* [Paper]
- [C2] A Roy, S Enam, J Xia, W.H Kim and V.S Lokhande. **Is Exchangeability better than I.I.D to handle Data Distribution Shifts while Pooling Data for Data-scarce Medical image segmentation?** *MIDL 2026* [Paper]
- [C3] A Roy, A Lee, R Chakraborty, V.S Lokhande. **ManifoldGD: Training-Free Hierarchical Manifold Guidance for Diffusion-Based Dataset Distillation.** *CVPR 2026* [Paper]
- [C4] P Dey*, A Roy*, H Cho, H.W Kim, V.S Lokhande. **MAGNET: Manifold-Aware Graph Diffusion Network for Connectome Generation.** *NeurIPS 2026*
- [C4] A Roy*, P Dey*, H Cho, H.W Kim, V.S Lokhande. **Privacy–Harmonization Tradeoffs in Data Space: Privacy-budgeted Pooling under Domain Shift.** *ICLR 2027* [Under Review]

Peer-Reviewed Journals and Magazines

- [J1] A Roy*, A.R Kaushik*, V.S Lokhande, N.K Ratha, V Govindaraju. **GE-FM: Geometry-aware Energy-based Flow Matching for Non Euclidean Manifolds.** *TMLR, 2026* [Paper]
- [J2] A Roy, D Faissol, A. P. Sales, R Chakraborty. **ADAPROTEINA: An Adapter Is All You Need For Developable Antibody Generation.** *TMLR, 2026* [Under Review]

Peer-Reviewed Workshop Papers and Extended Abstracts

- [W1] A Roy, T Hassan, R Ayyalasomayajula and V.S Lokhande. **Attending To Routers Aids Indoor Wireless Localization.** *AAAI ML4Wireless, 2025* [Paper]
- [A1] A Roy, WH Kim, R Zivadinov and V.S Lokhande. **Scanner-aware Causal Harmonization Of ADNI MRI for Alzheimer's Disease Detection.** *AAIC, 2026*

TEACHING EXPERIENCE

Graduate Teaching Assistant, State University at New York, Buffalo

CSE587: Data Intensive Computing, Fall 2024 CSE574: Introduction to Machine Learning, Spring 2025 CSE368: Introduction to Artificial Intelligence, Fall 2025 CSE573: Computer Vision and Image Processing, Spring 2026 CSE555: Introduction to Pattern Recognition, Fall 2026